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Abstract

Vibration-based condition identification of bolted connections can benefit the effective maintenance and operation of steel structures. Existing studies show that modal parameters are not sensitive to such damage as loss of preload. In contrast, structural responses in the time domain contain all the information regarding a structural system. Therefore, this study aims to exploit time-domain data directly for condition identification of bolted connection. Finite element (FE) model updating is carried out based on the vibration test data of a steel frame, with various combinations of bolts with loss of preload, representing different damage scenarios. It is shown that the match between the numerically simulated and measured acceleration responses of the steel frame cannot be achieved. The reason is that time-dependent nonlinearity is generated in bolted connections during dynamic excitation of the steel frame. To capture the nonlinearity, a virtual viscous damper is proposed. By using the proposed damper alongside the updated system matrices of the FE model, the time domain acceleration responses are estimated with great consistency with the measured responses. The results demonstrate that the proposed virtual damper is not only effective in estimating the time domain acceleration responses in each damage case, but also has the potential for condition identification of bolted connections with such small damage as just one bolt with loss of preload. It can also be applied to other challenging scenarios of condition identification, where modal parameters are not sensitive to the damage.

Keywords
Time domain condition identification, finite element model updating, non-linear damping, bolt pretension loss, non-linear constrained optimization

1 Introduction

Bolted steel connections are commonly used in steel construction. The bolts are usually pretensioned to increase the structural capacity of bolted connection. Loss of pretension force in the bolts occur over time due to viscoelastic creep, environmental effects and possible deterioration. Loss of pretension also occurs due to redistribution of stresses in the bolts due to weathering of materials in the surrounding steel members. It thus necessitates the use of condition monitoring of structures. By assessing the current condition, the current capacity of the structure can be evaluated. If the estimated capacity of the structure is found to be less than a threshold capacity, an indicator can be set for sanctioning repair or renovation of the structure.

Non-destructive testing tools have been extensively used in the existing literature[1–3] for identifying existing pretension force in the bolts. However, non-destructive testing techniques become very expensive for structures with many bolts. Finite element model updating[4] is commonly used for identifying condition of a structure by using global measurements taken along the length and breadth of the structure. Finite element model updating technique can be implemented on frequency domain or time domain measurements of the structural responses. For example, a virtual distortion method based model updating technique is proposed in the reference[5] for detecting damage in cable structures of a bridge. Further work is needed to implement the virtual distortion method for identifying rotational spring constants at the connections of a steel frame. Compared with the frequency domain data, structural vibration responses in time domain are more sensitive to damage[6;7], and is thus attracting increasing research interests. The accurate simulation of time-domain structural vibrations not only provides the opportunity for finite element model updating using time-domain data, but also can

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be a reliable approach to constructing data dictionary for machine-learning-based method. Finite element model updating is carried out for identifying the stiffness of bolted connections from the measured strain time histories[8]. Bilinear springs along with a Rayleigh damping matrix are used for the dynamic simulation of a pipe with bolted connection[9]. However, the results[9] show that the dynamic simulation can still be improved so that it matches closely with the measured responses.

Discontinuous interfaces at the bolted connections region introduce structural damping and nonlinear contact[10]. Micro and meso-scale parameters like geometry, roughness and bolt preload govern the contact nonlinearity[11], and it is computationally very expensive to include those effects in a macro-scale model. The study by Hammami et al.[12] concluded that the level of junction coupling has sufficient influence on the dynamic system to induce damping. It thus necessitates integrating the damping effects introduced at a bolted connection in the dynamic analysis of the structure.

A nonlinear system identification technique is employed for modelling and analysing the nonlinear damping effects induced at a bolted connection[13]. A nonlinear damping model is proposed by combining a viscous damping model with a quadratic polynomial damping force[14]. The nonlinear damping effects modelled in references[13;14] are applicable for a rigid mass system. A polynomial type nonlinear damping model is proposed for a rigid mass system[15], where the inherent non-linearity in the damping is estimated using wavelet transform. Matrix based and modal parameter based damping identification methods[16;17] are also reported. Damping identification based on matrix method requires the full frequency response function matrix to be measured. Modal parameter based damping identification requires prior knowledge about the damping matrix, which is again the sole objective of the method. A probability distribution of decay rate[18] is proposed for developing a relation between time domain damping to damage in a steel structure with bolted connections. However, it cannot quantify the effect of bolt loosening on the system matrices in the finite element model of the structure.

In order to integrate the nonlinear damping effects in the finite element modelling of structures with bolted connections, a virtual damper is proposed and introduced to the finite element model of the steel frame. It is used to capture nonlinear and time-dependent damping effects generated by the system which comprises the structure and

the impact device. In Section 2, the methodology is described in details, including the implementation of the virtual damper in the general dynamic equation of a structural system, and the finite element model updating technique. In Section 3, details of a steel frame with bolted connections designed and assembled in the laboratory is presented. Numerical modelling of the steel frame and the model updating results are given in Section 4. The numerical simulation results with and without the vitual damper have been compared with the measured acceleration responses for various damage cases, which demonstrate the effectiveness of the proposed method.

2 Methodology

2.1 Introduction of a virtual viscous damper to the dynamical system

The equation of motion of damped forced vibration of a structural system is given by

$$M\ddot{X} + C\dot{X} + KX = F + W_0 \tag{1}$$

where, M, C and K are structural mass, damping and stiffness matrices of the system, X is the system displacements, F is the applied force. W_0 is a Brownian motion[19] representing process noise that quantifies the uncertainties coming from modelling system dynamics. Without loss of generality, let the contribution of structural steel connection towards nonlinearity in the dynamic system response be expressed through a nonlinear damping matrix C^{NL}, such that $C^{NL} = C - C^L$. Here, C is the damping matrix of the structural system, taking into account the linear as well as nonlinear component of damping. The dynamic system can now be written as

$$M\ddot{X} + \left(C^L + C^{NL}\right)\dot{X} + KX = F + W_0 \tag{2}$$

where,

$$C^L = \alpha M + \beta K, \qquad and \quad C^{NL} = g_c\left(X, F\right) \tag{3}$$

The nonlinear component of damping C^{NL} is taken as a time varying function g_c of system states X and input force F. Moving the nonlinear component of damping C^{NL} to the right hand side, Equation 2 can be written as

$$M\ddot{X} + C^L\dot{X} + KX = F - C^{NL}\dot{X} + W_0 \tag{4}$$

or

$$M\ddot{X} + C^L\dot{X} + KX = F - F^d + W_0, \quad \text{where} \quad F^d = C^{NL}\dot{X} \tag{5}$$

For a given F^d, the dynamical system (Equation 5) can be solved as shown in Appendix A. Conversely, given the availability of a measured set of acceleration responses Z_m on the structure, F^d can be estimated through inverse problem solving process. A virtual viscous damper (F_h^d) is applied to the structural system as shown in Figure 1. The virtual

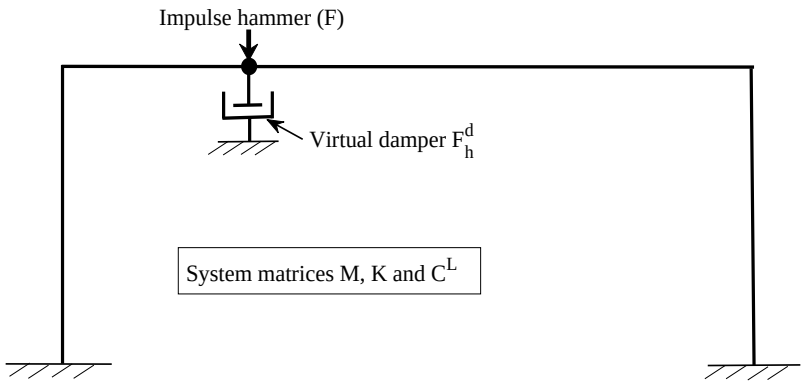


Figure 1. Virtual viscous damper.

damper with force F_h^d is placed exactly at the same location as the position where the impulse excitation is applied. It is assumed that by placing F_h^d at such a location will capture the nonlinear effects of the dynamical system including the uncertainties in the applied impulse excitation. The single unknown force corresponding to the virtual damper reduces number of unknown parameters to just one. Given uncertainties in the system and in the measurements, Metropolis Hastings algorithm[20] is used in this study to generate samples of F_h^d at every time step of the dynamical system.

2.2 The proposed condition identification approach

A two stage FE model updating process is proposed in this paper. In the first stage, the FE model is updated using the measured frequencies only. In the second stage, a virtual damper is introduced to the initial FE model, and the virtual damping force is estimated using the measured accelerations and impulse force in the time domain. A summary of all the steps involved in condition identification of a steel frame with bolted connections is shown in Figure 2, and are described here. In step 1, the dynamic system

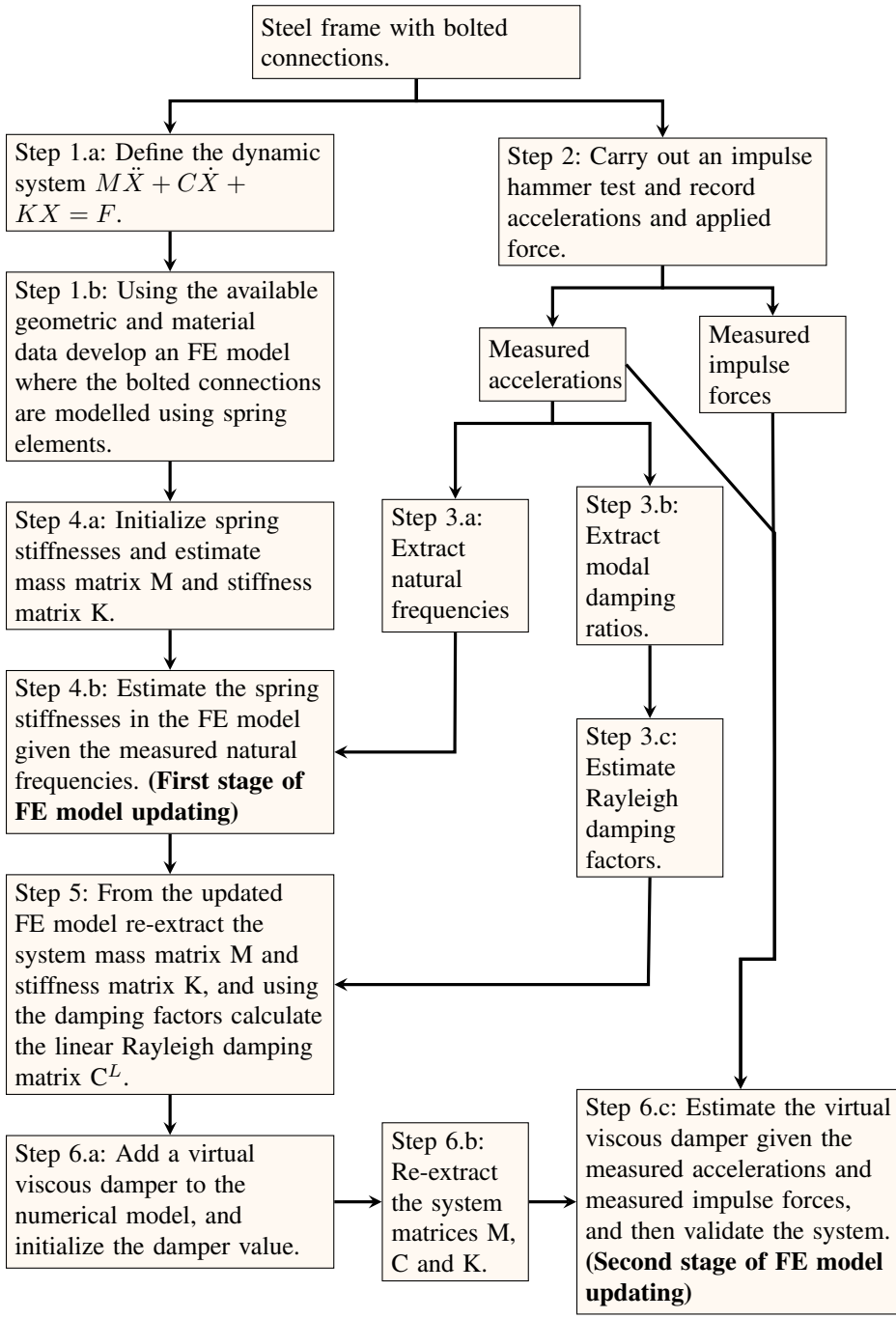


Figure 2. Steps involved in experimental identification of steel frames with bolted connections

is defined, and the finite element model of the steel frame is constructed based on the available geometry details and nominal material properties. In step 2, the modal test is carried out, and the accelerations and the applied impulse force are recorded. In step 3, Natural frequencies and the modal damping factors are extracted from the measured acceleration time histories. The Rayleigh damping factors are then evaluated using the measured modal damping factors and natural frequencies. In step 4, the beam column joint of the steel frame are modelled using linear spring elements. A differential evolution algorithm is then implemented to update the initial finite element model by estimating the stiffness of all the springs using the measured natural frequencies of the steel frame. In step 5, Rayleigh damping matrix is constructed using the the system mass and stiffness matrices extracted from the updated finite element model and the Rayleigh damping factors evaluated in step 3. The measured impulse force is then applied to the system, and the dynamic analysis of the system is carried out. The simulated acceleration responses of the system are then compared with the measured acceleration responses. In step 6, the proposed virtual damper is introduced to the updated finite element model of the steel frame. The system matrices are re-generated by incorporating the virtual damper into the system. Metropolis Hastings algorithm is then used to estimate the virtual damping force so that the estimated acceleration time histories from the updated finite element model match closely with the measured acceleration responses. The above procedure is then validated with ten different damage scenarios.

2.3 Finite element model updating for estimating spring stiffnesses given the measured natural frequencies

Finite element model updating technique[4] employed in Step 4.a (first stage of model updating) is an inverse problem that estimates the unknown spring stiffnesses in the FE model by minimising the cumulative percentile error between the measured and simulated frequencies. Let the vector of updating parameters θ in the finite element model updating be the multipliers to the initial spring stiffness, and are defined as $\theta = \{Kms_i\}_{i=1,2,\ldots,6}$. The ith element of θ is defined as $Kms_i = Kf_i/k_i$, where k_i denotes the stiffness of ith spring before model updating and Kf_i denotes the stiffness of ith spring after model updating. An objective function $g(\theta)$ is formed that minimises the sum of square of error between the measured and simulated frequencies. If f_m is the vector of measured frequencies and $f_s(\theta)$ is the vector of simulated frequencies for a

given set of parameters of interest θ, the objective function $g(\theta)$ is defined as

$$g(\theta) = Err^T \times Err, \qquad \text{where, } Err = f_m - f_s(\theta) \tag{6}$$

For a given candidate solution $\hat{\theta}$ of the unknown parameter θ, multipliers Kms to the initial stiffness Kf are obtained as $Kms = \hat{\theta}$. Stiffness of ith spring in the finite element model is then calculated as $k_i = Kms_i \times Kf_i | i = 1, 2, ..., 6$. A frequency analysis is then carried out in the finite element software OpenSEES [21;22] to estimate the natural frequencies of the system f_s using the updates spring stiffensses $k_{i|i=1,2,...,6}$ for a given candidate solution $\hat{\theta}$ of the unknown parameter θ.

The finite element model updating technique implemented in this study involves finding the optimal solution of the objective function (Equation 6). In this paper a differential evolution (DE) algorithm [23] was used to find the optimal values of the parameters (θ) of the finite element model given the measured frequencies (f_m) of the structural system. The DE algorithm is one of the most stable and versatile population based metaheuristic search algorithm [24]. The DE algorithm has shown robustness to multi-modal nonlinear optimization problems by iteratively improving a candidate solution based on an evolutionary process. The details of the DE algorithm are given in the paper by Storn and Price [23].

2.4 Metropolis Hastings algorithm for generating samples from $p\left(F_h^d | Z_m\right)$

Step 6.c in Figure 2 (second stage of model updating) is an inverse problem process of generating samples form the target probability distribution $p\left(F_h^d | Z_m\right)$ given the output measurement data Z_m. While, $p(F_h^d)$ is the prior density function of the uncertain parameter F_h^d, and $p(Z_m | F_h^d)$ is the likelihood of measurement with respect to the given parameter F_h^d, the steps involved in implementing the Metropolis Hastings algorithm [20] are

1. Start with an initial parameter $F_h^{d,0}$
2. Generate a new parameter $F_h^{d,'}$ from proposal density function $q(F_h^d | F_h^{d,n})$
3. Accept $F_h^{d,'}$ with probability α^p, defined as

$$\alpha^p(F_h^{d,'} | F_h^{d,n}) = \min \left\{ 1, \frac{p(Z_m | F_h^{d,'}) p(F_h^{d,'}) q(F_h^{d,n} | F_h^{d,'})}{p(Z_m | F_h^{d,n}) p(F_h^{d,n}) q(F_h^{d,'} | F_h^{d,n})} \right\}$$

4. Repeat steps 2 to 5 till convergence or the maximum number of samples covered

Convergence of the process is achieved when a Markov chain reaches a stage such that the posterior density function remains unchanged (i.e. variance of the last three generated sample is less than 0.01) for all subsequent stages. If the maximum number of samples is covered before convergence is achieved, the maximum number of samples is increased and the algorithm is run again. In this paper the likelihood is taken as a multivariate normal distribution. Each element of the measurement vector is taken as an independent and normally distributed random variable with mean as the measured value and standard deviation as 2% of mean.

3 Experimental test set-up and modal parameter extraction

3.1 Test set-up

In the present study a single bay single storey steel portal frame with bolted connections is designed and manufactured in the laboratory. The beam is connected to the columns with the help of gusset angles and pre-tensioned bolts. Bright zinc plated high tensile 10 mm bolts of grade 8.8 are used. The maximum recommended tightening torque[25] for 10 mm bolts of grade 8.8 is 55 Nm. A pretension torque of 55 Nm is applied to each bolt using a torque wrench. The columns are welded to the base plates which are then bolted to the strong floor using 16 mm bolts. All the bolts used in this study are high tensile bolts of grade 8.8. The geometric details of the frame and bolted connection details with specified bolt numbers are shown in Figure.3. In this study, ten damage cases are considered. Damage is introduced in the portal frame by loosening pretensioned bolts. Care was taken to ensure that the bolts are only loosened and not removed from the connections. Details of the ten damage cases are given in Table.1. The objective of the present study is to identify the condition of the structural connections, given the occurrence of a damage scenario out of the ten cases mentioned in Table.1. More details about the experiment and its results can be found in[7].

3.2 Natural frequencies

The natural frequencies of the steel frame were extracted from the measured acceleration time signals using peak-picking technique. The extracted natural frequencies are given in Table 2.

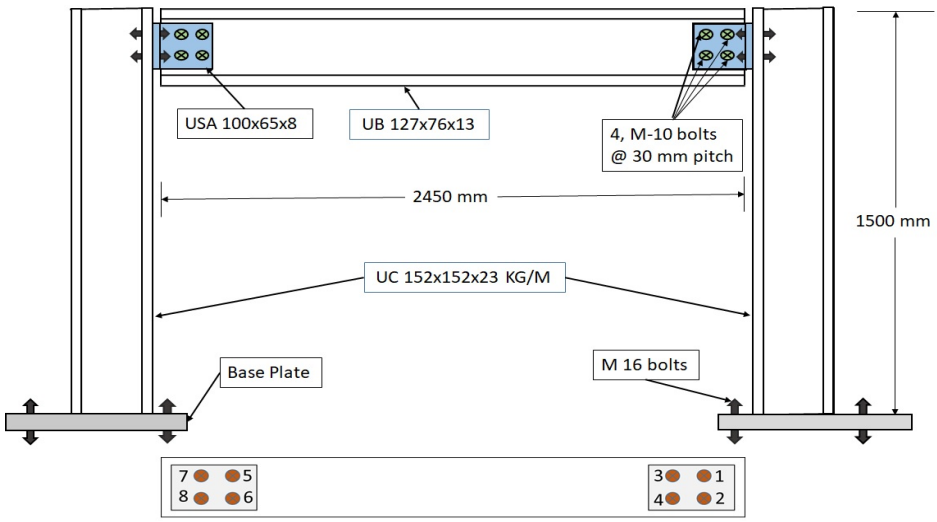


Figure 3. Details of steel members and numbering of bolts.

Table 1. Description of damage cases.

Description	Damage cases	Bolt loosened
Intact	D0	All tight
One bolt loosened at one end and different locations	D1	1
	D2	3
	D3	2
Two bolts loosened at one end and different locations	D4	1 and 2
	D5	1 and3
	D6	1 and 4
Two bolts loosened at both ends and different locations	D7	1, 2, 5 and 6
	D8	1, 3, 5 and 7
Three bolts loosened at one end	D9	1, 2, and 3
Four bolts loosened at one end	D10	1, 2, 3 and 4

3.3 Modal damping factor estimation

Output-only modal identification techniques[26;27] have shown great potential in identifying modal parameters from measured time domain responses of a structural system. Modal damping factors are estimated from the measured time signals using the time domain decomposition technique[26]. The time domain decomposition technique (TDD) is implemented in this study, because of (i) its simplicity and efficiency, and (ii) possibility of extracting modal damping factors given any arbitrary input. The TDD technique for modal damping factor extraction can be described in four steps (i) the

Table 2. Natural frequencies.

Damage scenario	Measured natural frequencies in Hz		
	1_{st}	2_{nd}	3_{rd}
0	82.430	258.800	563.200
1	83.340	254.300	563.300
2	83.340	254.700	564.600
3	82.920	254.700	563.300
4	81.630	251.700	560.700
5	82.920	254.700	562.500
6	83.340	254.300	563.700
7	81.630	251.700	560.700
8	82.920	255.200	561.600
9	81.210	250.900	559.000
10	78.640	245.300	550.100

prior information of the ranges of the frequencies are used along with a digital band-pass filter to isolate filtered time signals Y_i corresponding to each mode, where i is the mode number, (ii) the energy correlation E_i of the ith mode is then calculated as $E_i = Y_i^T \times Y_i$, (iii) singular value decomposition of energy correlation matrix, defined as $E_i = U \times \Omega \times U^T$, gives the singular vector matrix U, the first column vector of which is designated as the mode shape of that isolated mode, and (iv) half-power band method is then used to extract damping factors from the auto power spectrum of the isolated time signals corresponding to each mode.

3.4 Rayleigh damping coefficients estimation

The estimated Rayleigh damping coefficients using the modal damping factors extracted in Appendix B are given in Table 3.

Table 3. Rayleigh damping coefficients for different damage cases.

Damage scenario	Modal damping factors			Rayleigh damping coefficients	
	1_{st}	2_{nd}	3_{rd}	α	β
0	0.0015	0.0018	0.0016	1.463	9.293e-7
1	0.0015	0.0044	0.0018	1.9	1.477e-6
2	0.0016	0.0047	0.0014	2.148	1.311e-6
3	0.0016	0.0050	0.0017	2.116	1.521e-6
4	0.0017	0.0052	0.0020	2.158	1.701e-6
5	0.0016	0.0049	0.0018	2.074	1.557e-6
6	0.0015	0.0044	0.0018	1.9	1.475e-6
7	0.0016	0.0048	0.0016	2.077	1.44e-6
8	0.0017	0.0048	0.0021	2.084	1.683e-6
9	0.0017	0.0073	0.0022	2.462	2.176e-6
10	0.0089	0.0211	0.0023	11.613	3.769e-6

4 Dynamic response in the presence of virtual damper

4.1 Numerical modelling of connection stiffnesses

OpenSEES[21;22] is used in this study for finite element modelling of the steel frame. A total of 33 nodes are assigned to the finite element model as shown in Figure 4. Eleven equally spaced nodes are present in the beam and columns. 'elasticBeamColumn'

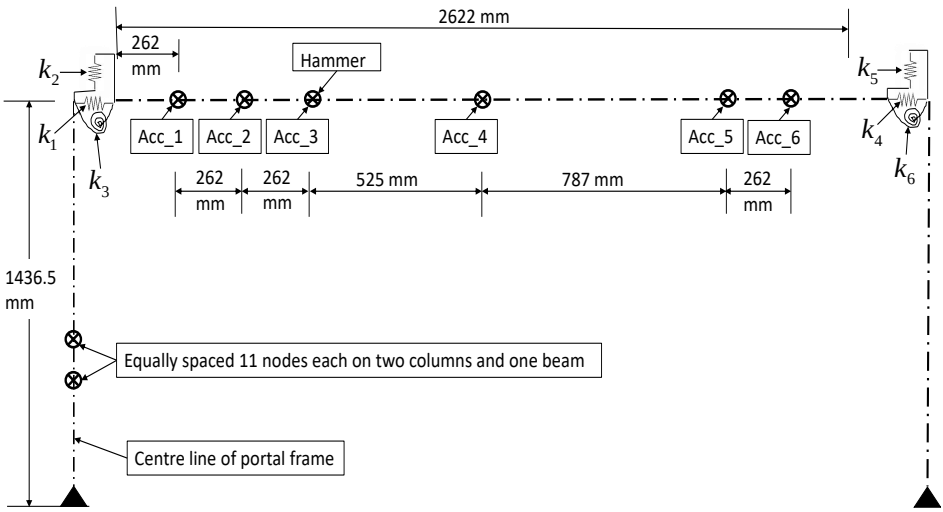


Figure 4. Finite element model in OpenSEES showing accelerometers and impulse hammer locations based on damage measurability.

elements in OpenSEES are used to model the beam and columns. Each beam column joint is modelled by three springs, two translational springs and one rotational spring (Figure 4). Let k_1, k_2 and k_3 be the spring stiffness in the left side joint, and k_4, k_5 and k_6 be the spring stiffness in the right side joint. As shown in Figure. 4, k_1 and k_4 act along the length of beam, k_2 and k_5 act transverse to the length of beam, and k_3 and k_6 act along the rotational degrees of freedom between the beam and column. In OpenSEES, 'zeroLength' elements are used to model the springs. According to Eurocode 3[28], such joints can be treated as semi-rigid joints, which allow a certain amount of rotation. For a beam column joint, where the elastic rigidities of the beam and column are of the same order, the relationship between rotational spring stiffness S and the fixity factor α_s can be written[29] as

$$\alpha_s = \frac{1}{1+\dfrac{3EI}{LS}} \quad \text{or} \quad S = \frac{3EI}{L} \times \frac{\alpha_s}{1-\alpha_s} \tag{7}$$

where, E is the elastic modulus of beam, I is second moment of inertia of beam and L is length of beam. The fixity factor α_s depends on the type of joints[8;28], and for a semi-rigid joint α_s varies between 0.143 to 0.891. Assuming a fixity factor of 0.6, the rotational spring stiffnesses k_3 and k_6 are each calculated as 1136498 Nm/rad. According to Eurocode 3[28], the translational stiffness of joints with more than two columns of preloaded bolts can be taken as an infinitely large number. In the finite element model the stiffness of each spring k_1, k_2, k_4 and k_5 is assumed to be 1×10^{10} N/m.

Six accelerometers are used in this study to measure the dynamic behaviour of the steel frame when the steel frame is excited with an impulse hammer. The optimal location of accelerometers and position of impulse excitation are evaluated based on the optimal sensor positioning algorithm given in[30]. The first three natural frequencies corresponding to the bending modes simulated for the damage case D0 are 85.23 Hz, 273.61 Hz and 566.27 Hz respectively. The cumulative percentile error between the measured (Table 2) and simulated frequencies is 9.7%.

Differential evolution algorithm[23] is implemented in Python to find the unknown parameters (multipliers to the initial spring stiffness) as the roots of the objective function given in Equation. 6. The estimated multipliers to the initial stiffness of the springs for the damaged case D0 are $Kms_1 = 0.0026$, $Kms_2 = 8.9204$, $Kms_3 = 0.2958$, $Kms_4 = 7.3604$, $Kms_5 = 8.9314$, and $Kms_6 = 1.5821$. By multiplying the initial spring stiffness to the corresponding stiffness multipliers, the final values of the spring stiffness for damage case D0 are given in Table 4. The estimated

Table 4. Updated spring stiffness for damage case D0.

Damage scenario	Spring stiffnesses					
	Kf_1, N/m	Kf_2, N/m	Kf_3, Nm/rad	Kf_4, N/m	Kf_5, N/m	Kf_6, Nm/rad
0	2.59×10^{7}	8.92×10^{10}	3.36×10^{5}	7.36×10^{10}	8.93×10^{10}	1.79×10^{6}

vertical translation and rotation spring stiffness at the two connections were observed to have different values, even though for damage case D0 they are normally considered identical. The difference in the vertical translation and rotation spring stiffness at the two connections can be attributed to the possible imperfections in the manufacturing and assemblage of the steel portal frame. It is clear that stiffness of springs at left side connection are less than the respective stiffnesses at the right side connection. To verify

the cause for this difference, additional accelerometers were installed on both columns of the portal frame. The mode shapes of the frame for the first three modes are given in Appendix C. The asymmetry behaviour of the frame is evident in Figures C.1-C.3. It can be seen that the beam-to-column interaction is markedly different at the two ends of the beam. Due to the observed asymmetry in the connections, the sway displacement at top of the left column is larger than that of its counterpart on the right. This is reflected in the updated stiffness parameters at the two ends, as can be seen in Table 4 in the paper.

The initial stiffness for damage cases D1 to D10 are taken same as the final stiffness for damage case D0. The estimated multipliers to the initial stiffness for damage cases D1 to D10 are given in Table 5 for damage cases 1 to 10. The comparison between

Table 5. Estimated multipliers to initial spring stiffness in the finite element model.

Damage scenario	Multipliers to spring stiffnesses					
	Kms_1	Kms_2	Kms_3	Kms_4	Kms_5	Kms_6
1	1.0000	1.0000	1.0000	0.9457	1.0000	1.0000
2	1.0000	1.0000	1.0000	0.9505	1.0000	1.0000
3	0.9984	0.9998	1.0000	0.9504	0.9988	0.9997
4	0.5320	0.8675	0.9046	0.9177	0.7649	0.2803
5	0.9965	0.9964	1.0000	0.9504	0.8995	0.9999
6	0.9922	0.9995	1.0000	0.9457	0.9973	0.9999
7	0.5078	0.8809	0.9052	0.9177	0.7244	0.2023
8	0.9831	0.9973	1.0000	0.9563	0.5000	0.9995
9	0.3692	0.7815	0.8586	0.9103	0.5811	0.1674
10	0.2499	0.4402	0.6202	0.8506	0.2427	0.1155

simulated frequencies using updated stiffness of springs in the finite element model with the measured frequencies is shown in Table 6. From the results shown in Table 6, the

Table 6. Comparison of estimated to measured natural frequencies.

Damage scenario	Measured natural frequencies in Hz			Simulated natural frequencies in Hz			Cumulative error in %
	1_{st}	2_{nd}	3_{rd}	1_{st}	2_{nd}	3_{rd}	
0	82.430	258.800	563.200	82.430	258.800	563.198	3.551e-4
1	83.340	254.300	563.300	82.430	254.299	563.052	1.136
2	83.340	254.700	564.600	82.430	254.699	563.065	1.364
3	82.920	254.700	563.300	82.430	254.696	563.064	0.634
4	81.630	251.700	560.700	81.630	251.700	560.700	0.000
5	82.920	254.700	562.500	82.427	254.698	562.943	0.516
6	83.340	254.300	563.700	82.429	254.298	563.045	1.210
7	81.630	251.700	560.700	81.631	251.713	560.731	0.011
8	82.920	255.200	561.600	82.409	255.195	562.047	0.538
9	81.210	250.900	559.000	81.214	250.937	559.120	0.041
10	78.640	245.300	550.100	78.640	245.300	550.099	1.817e-4

maximum cumulative percentile error for the three simulated frequencies is 1.364%. It is possible to quantify damage at the connections using the estimated spring stiffness corresponding to different damage cases. A damage indicator can be proposed to quantify damage at the connections based on the changes in the estimated spring stiffness values. The possible damage indicators will be explored in a future study. The finite element model with the updated stiffnesses of springs are used to extract the structural system matrices. In OpenSEES[21] a 'Static' analysis with 'FullGeneral' system is performed to extract the full stiffness matrix, and a 'Transient' analysis with 'FullGeneral' system is performed to extract the full mass matrix. The damping matrix is estimated in the next section, which is then used for the dynamical analysis of the steel frame with bolted connection.

4.2 Dynamic response without virtual damper

Using the structural stiffness matrix (K) and mass matrix (M) of the steel frame extracted from OpenSEES in Section 4.1, and the the Rayleigh damping coefficients α and β extracted in Section 3.4, the linear damping matrix (C^L) is calculated as shown in Equation. 3. Let F be the vector of forces applied to the steel frame. All the elements of vector F are zero except at the degree of freedom where the impulse is applied. A measured force from impulse hammer test is shown in Figure 5. Let Z_m be the vector

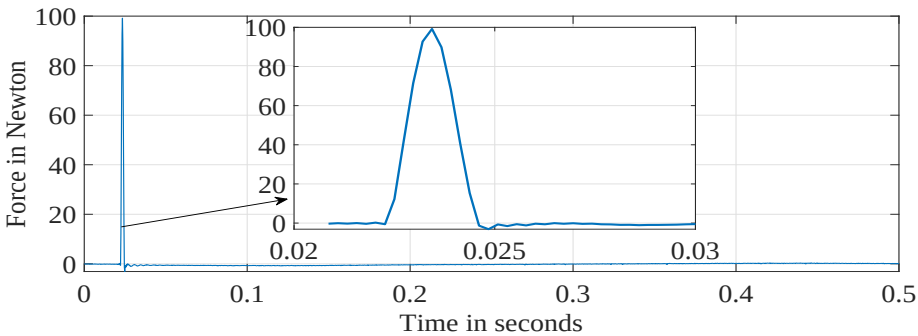


Figure 5. Measured force from impulse hammer test.

of measured accelerations. Six accelerometers are used in this study, so the size of Z_m is 6×1. Since the finite element model (Figure 4) has 33 nodes, two nodes are fixed, and every node has three degrees of freedom (two translation and one rotation), the size of stiffness matrix K is 93×93. The size of Y^{n+1} in Equation 14 is 186×1. If Z_e be the vector of estimated acceleration responses at the same degrees of freedom as the

measured accelerations Z_m, at time step $n+1$, Z_e is calculated as

$$Z_e^{n+1} = H \times Y^{n+1} \tag{8}$$

where, H is a rectangular sparse matrix with number of rows is equal to number of measurements, and number of columns is equal to the number of degrees of freedom of the structural system (e.g. length of stiffness matrix). Let nm is the number of measurements and nd is the number of degrees of freedom of the system. Then the size of the matrix H is $nm \times nd$, and elements of H are defined as

$$H_{i,j} = \begin{cases} 1 & \text{for measurement number } i \text{ and the corresponding degree of freedom } j \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

If the first measurement is taken along the nf^{th} degree of freedom, then $H_{1,nf} = 1$, and all the elements in the first row of matrix H are zeros. Here the size of matrix H is 6×186. The estimated acceleration responses Z_e are plotted against the measured responses Z_m in Figure 6.

From Figure 6, it is clear that apart from the acceleration at location 4, the estimated accelerations at all other locations do not match closely to the measured accelerations at those locations. Even though the updated finite element model has been used along with Rayleigh damping coefficients generated from the measured modal damping factors, a big difference between the estimated and the measured responses can be seen in Figure 6. As reported in [13–15], the nonlinear damping effects generated from the structural connections must be taken into account in the dynamic system equation for analysing the dynamic behaviour of a structure.

4.3 Dynamic analysis of steel frame in the presence of the virtual damper

The external virtual damping force F_h^d is estimated following the procedure described in Section 2. The estimated damping force for damage case 0 is shown in Figure 7. In Equation 5, at degree of freedom where the impulse is applied, the element of F^d is replaced with the estimated damping force F_h^d. By keeping all other elements of F^d as zero, Equation 5 is then solved as in Equation 14. The estimated acceleration responses are then compared with the measured accelerations at the respective locations and are

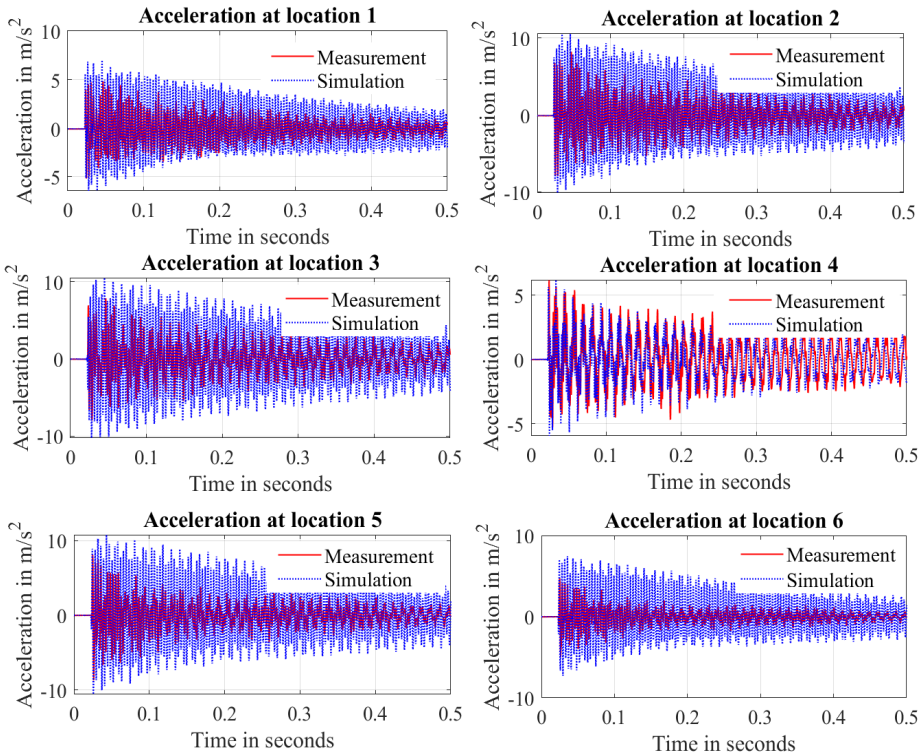


Figure 6. Comparison of acceleration responses.

shown in Figure 8. To quantify the difference between the measured and simulated accelerations the average power is calculated between measured accelerations, and simulated accelerations before and after damping update. If $\tilde{x}(t)$ be a time signal, the average power $P_{\tilde{x}}(t)$ between time t_1 and t_2 is given by

$$P_{\tilde{x}}(t) = \int_{t_1}^{t_2} |\tilde{x}(t)|^2 \, dt \tag{10}$$

The percentile absolute change in average power between measured and simulated accelerations are given in Table 7. From the results shown in Table 7 it is clear that the damping update substantially reduces the difference in average power between the measured and simulated accelerations. The mean of difference in average power corresponding to the six locations of accelerometers is 874.98 in the absence of virtual

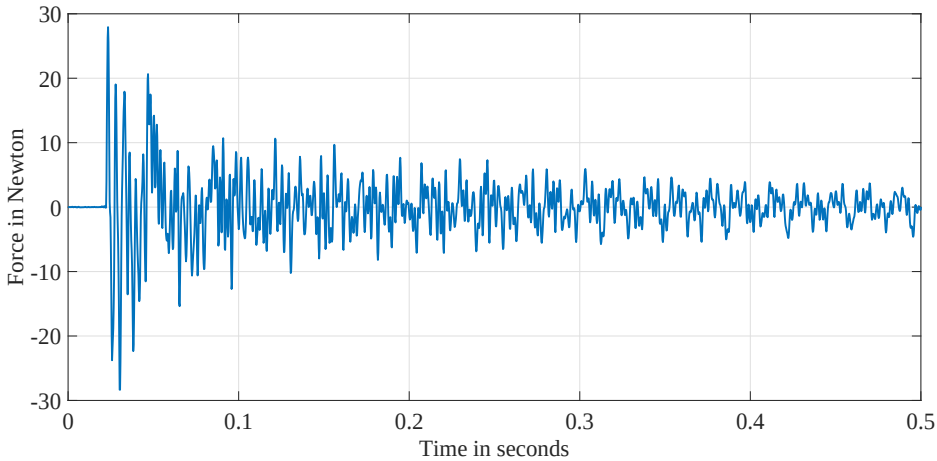


Figure 7. Estimated damping force.

damper, whereas in the presence of virtual damper the mean of difference in average power reduces to 8.25.

Table 7. Percentage change in average power.

Updating scenario	Accelerometer locations					
	1	2	3	4	5	6
Without virtual damper	492.02	438.92	331.18	25.27	544.22	418.27
With virtual damper	5.37	6.54	12.43	1.40	14.97	8.79

The frequency spectrum of measured accelerations at location 1 were compared to the frequency spectrum of simulated accelerations with and without virtual damper, and is shown in Figure 9. It is clear from Figure 9 that the proposed technique not only estimates accelerations that compare well in the time domain, the estimated accelerations also compare well to those of measured accelerations in the frequency domain. It is also clear that there is a large deviation between the frequency spectrum of measured acceleration to the frequency spectrum of simulated acceleration without the virtual damper, and thus highlights the importance of the introduction of the proposed virtual damper to the dynamical system.

The comparison of estimated damping forces corresponding to various damage scenarios both in the time domain and frequency domain is given in Appendix D. As shown in Figures D.1-D.2, variation of the estimated damping force among various damage scenarios is very small apart from the damage scenario 10, where all the four

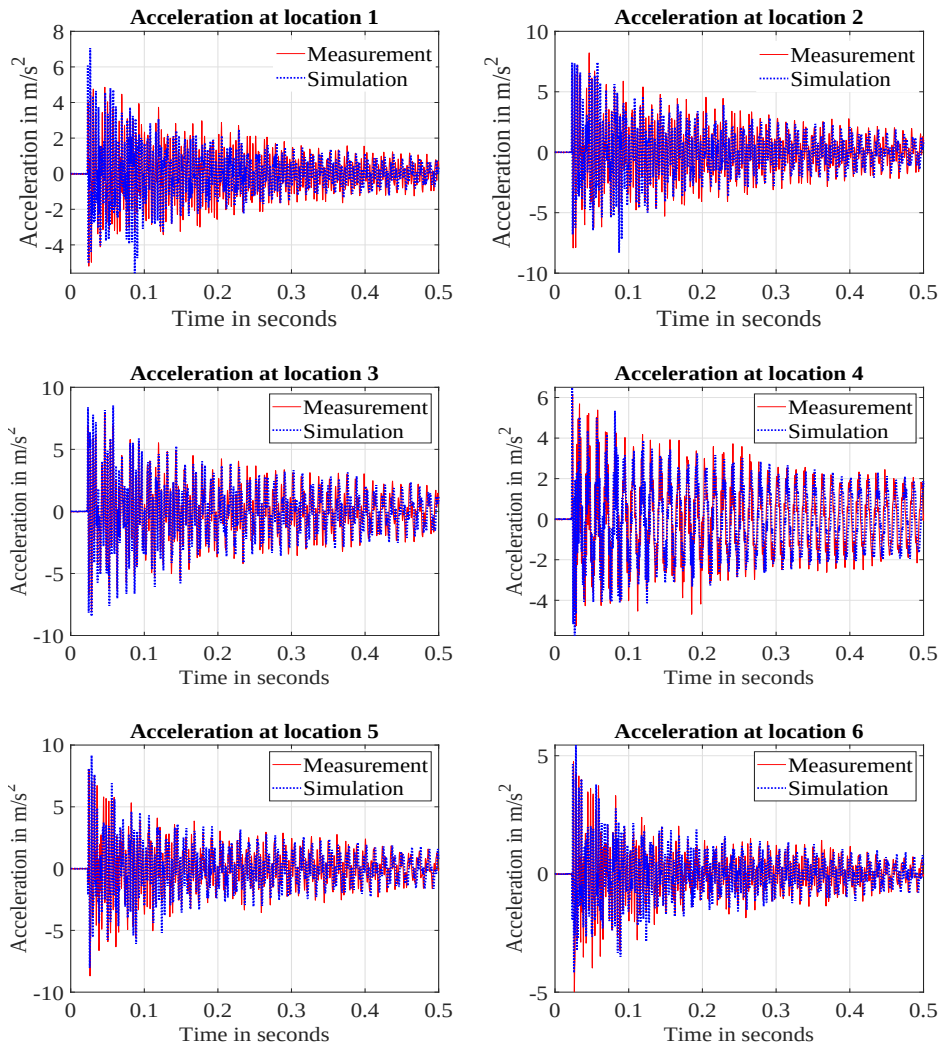


Figure 8. Comparison of acceleration responses for damage case 0 after damping update.

bolts at the right side connection were loosened. In damage scenario 10, this could be attributed to the total loss of friction forces between the connection angles and the web of the beam by loosening all the four bolts at the connection. For completeness, the comparison of estimated acceleration responses for all considered damage scenarios is given in Appendix E.

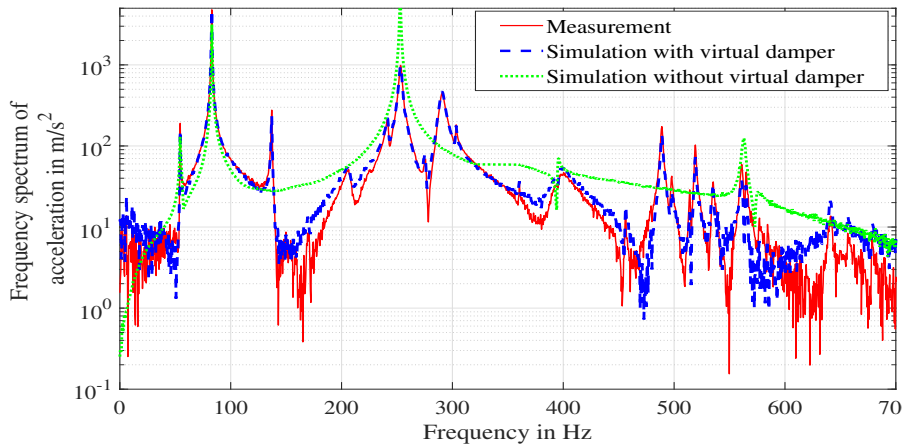


Figure 9. Frequency spectrum of acceleration at location 1.

In the paper by Cao et al.[31], it has been reported that nonlinear damping force generated due to structural connection is dependent on the amplitude of impulse excitation. To evaluate the effect of amplitude of impulse excitation on the estimated damping force, the steel frame is excited with 10 excitations of varying amplitude, for each damage case. The amplitude of the ten applied impulse excitations for damage case 1 are shown in Figure 10. The corresponding estimated damping forces are shown in Figure 11. It

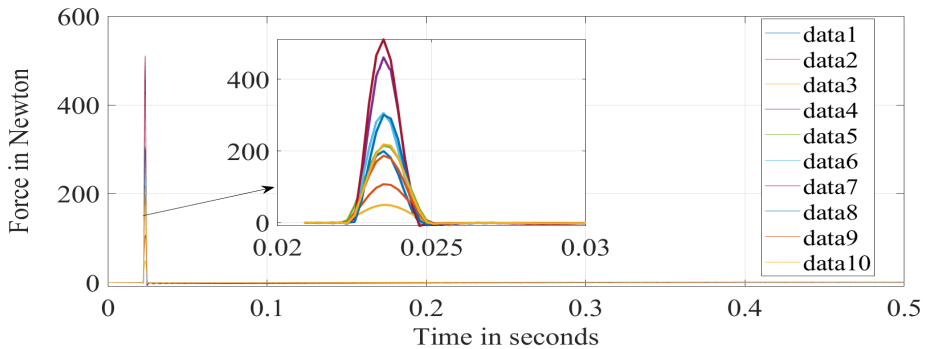


Figure 10. Multiple impulse excitations of varying amplitude for damage case 1.

can be noted from Figure 11 that even though the amplitudes of estimated damping forces are different for different amplitude of applied impulse excitations, the estimated damping forces are shown to follow a similar pattern. This can be very beneficial in

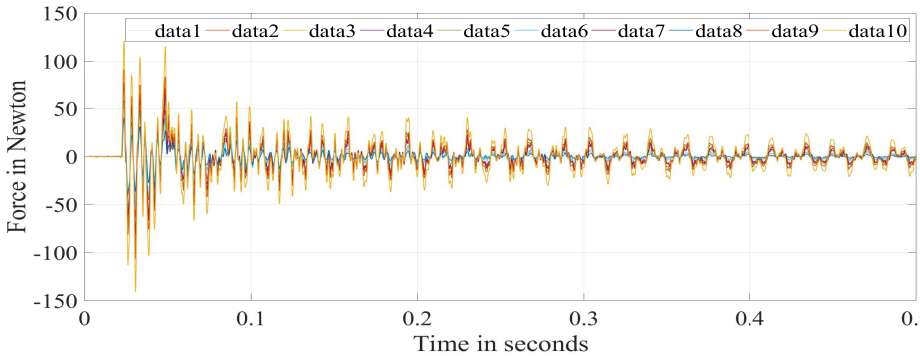


Figure 11. Estimated damping force for multiple impulse excitations of varying amplitude.

developing damage classification algorithms based on the measured time signals. A future study is focussed on developing machine learning tools for identifying damages, where a set of neural networks are being trained using the estimated damping forces corresponding to various damage cases.

To get a more qualitative explanation and a better visual understanding of the dependence of amplitude of excitation on estimated nonlinear damping force, the measured accelerations are normalized with respect to the maximum amplitude of applied impulse excitations. All the measured accelerations are normalized with reference to a 100 Newton of maximum amplitude for the ten applied impulse excitations. The estimated damping forces using the normalized measured accelerations for all the damage cases are shown in Figure 12. As shown in Figure 12, for a given damage case, the estimated damping forces not only follow the same pattern but also the amplitudes to a great extent. It thus can be stated that there is a linear relationship between the amplitude of applied impulse excitation to that of the estimated damping force.

5 Conclusion

Assessment of the current condition of a steel frame with different combinations of loosening of bolts in the structural connections has been carried out in this study. Numerical modelling of the steel frame is performed in OpenSEES. The stiffness of the finite element model is updated using measured natural frequencies so that the cumulative percentile error in the simulated frequencies is less than 2%. The estimated system matrices predicts the current condition of a structural connection as a single unit,

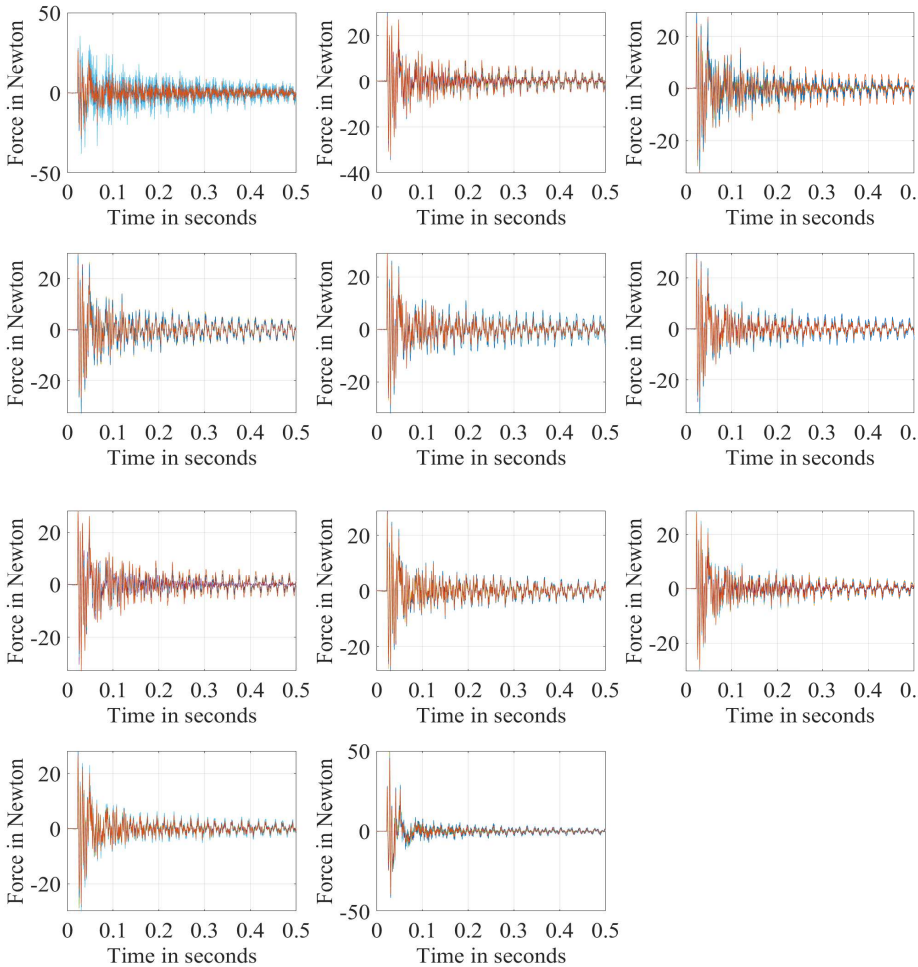


Figure 12. Comparison of estimated damping force for all damage cases across 10 impulse excitations.

represented by mechanical springs. A virtual damper to the steel frame is proposed for identifying the system in time domain. The virtual damper generates a time varying force which when applied to the dynamical system gives very good agreement between the measured accelerations and accelerations simulated from the finite element model of the steel frame in all damage cases and in both time and frequency domains.

From the results it can be seen that the estimated damping forces have a linear relationship to the amplitude of applied impulse excitations. The observed linear relationship can be advantageous in condition identification and damage detection in structures using output only methods or techniques based on machine learning where the measured data is usually normalized for the decision making process. The estimated damping forces for different damage cases can be taken as input data variables in an ongoing project for classifying damage cases based on output only measured accelerations.

Acknowledgments

The authors would like to acknowledge Engineering and Physical Sciences Research Council, UK Research and Innovation, for the financial support with Grant Reference Number: EP/R021090/1.

A Solution of stochastic differential equation 1

The equation of motion of damped forced vibration (Equation 1) is given by

$$M\ddot{X} + C\dot{X} + KX = F + W_0 \tag{11}$$

$$\begin{aligned} &\text{Let, } X_1 = X, \quad \text{and} \quad X_2 = \dot{X} \; or \quad dX_1 = X_2 dt \\ &\text{Then, } \dot{X_2} = \ddot{X} = M^{-1}F - M^{-1}C\dot{X} - M^{-1}KX + W_1, where\ W_1 = M^{-1}W_0 \\ &or\ dX_2 = \left[M^{-1}F - M^{-1}C\dot{X} - M^{-1}KX\right] dt + \sigma dW \end{aligned} \tag{12}$$

Putting in a vector form

$$\begin{aligned} &\begin{Bmatrix} dX_1 \\ dX_2 \end{Bmatrix} = \begin{Bmatrix} X_2 \\ M^{-1}F - M^{-1}C\dot{X} - M^{-1}KX \end{Bmatrix} dt + \begin{Bmatrix} 0 \\ \sigma \end{Bmatrix} dW \\ &or\ dX = \mu_X dt + \sigma_x dW \end{aligned} \tag{13}$$

The drift term (μ_X) in the stochastic differential equation (Equation 13) is of order 2, and the diffusion term (σ_X) in Equation 13 is of order 1.5. Order 1.5 strong Taylor expansion[19] can be used to solve Equation 13. An implicit scheme is required to avoid instability when a large step size is used for time integration. Using implicit order 1.5 strong Taylor scheme[19], the solution of above stochastic differential equation (Equation 13) can be written as

$$Y^{n+1} = AY^n + BU^n + DR \tag{14}$$

where,

$$Y = \begin{Bmatrix} X_1^{n+1} \\ X_2^{n+1} \end{Bmatrix} \tag{15}$$

$$A = \begin{bmatrix} I - \frac{1}{2}\Delta^2 M^{-1}K & \Delta I - \frac{1}{2}\Delta^2 M^{-1}C \\ -\Delta M^{-1}K + \frac{1}{2}\Delta^2 M^{-1}CM^{-1}K & I - \Delta M^{-1}C - \frac{1}{2}\Delta^2 M^{-1}K + \frac{1}{2}\Delta^2 M^{-1}CM^{-1}C \end{bmatrix} \tag{16}$$

$$B = \begin{bmatrix} \frac{1}{2}\Delta^2 M^{-1} & 0 \\ \Delta M^{-1} - \frac{1}{2}\Delta^2 M^{-1}CM^{-1} & \frac{1}{2}\Delta^2 M^{-1} \end{bmatrix} \tag{17}$$

$$U = \begin{Bmatrix} F \\ \dot{F} \end{Bmatrix} \tag{18}$$

$$D = \begin{bmatrix} 0 & \sigma \\ \sigma & -M^{-1}C\sigma \end{bmatrix} \tag{19}$$

and

$$R = \begin{Bmatrix} \Delta W \\ \Delta Z \end{Bmatrix} \tag{20}$$

In the present study ΔW and ΔZ are generated from a multivariate normal distribution with mean R_{mean} and covariance matrix R_{cov}, where

$$R_{mean} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \text{and} \quad R_{cov} = \begin{bmatrix} \Delta & \frac{1}{2}\Delta^2 \\ \frac{1}{2}\Delta^2 & \frac{1}{3}\Delta^3 \end{bmatrix} \tag{21}$$

B Rayleigh damping coefficients from measured modal damping ratios

Using Rayleigh's proportional damping, the modal damping factors can be written as

$$\zeta_i = \frac{1}{2\omega_i} \times \alpha + \frac{\omega_i}{2} \times \beta \tag{22}$$

The Rayleigh damping coefficients are then estimated using least square sense as

$$\begin{Bmatrix} \zeta_1 \\ \zeta_2 \\ \vdots \\ \zeta_n \end{Bmatrix} = \begin{bmatrix} \frac{1}{2\omega_1} & \frac{\omega_1}{2} \\ \frac{1}{2\omega_1} & \frac{\omega_i}{2} \\ \vdots & \vdots \\ \frac{1}{2\omega_1} & \frac{\omega_i}{2} \end{bmatrix} \times \begin{Bmatrix} \alpha \\ \beta \end{Bmatrix}, \; or \, \{\zeta\} = [A_w] \times \begin{Bmatrix} \alpha \\ \beta \end{Bmatrix}, \; where \, [A_w] = \begin{bmatrix} \frac{1}{2\omega_1} & \frac{\omega_1}{2} \\ \frac{1}{2\omega_1} & \frac{\omega_i}{2} \\ \vdots & \vdots \\ \frac{1}{2\omega_1} & \frac{\omega_i}{2} \end{bmatrix} \tag{23}$$

Then

$$\begin{Bmatrix} \alpha \\ \beta \end{Bmatrix} = \left[A_w^T \times A_w\right]^{-1} \times \left[A_w^T\right] \times \zeta \tag{24}$$

C Normalized mode shapes of the portal frame

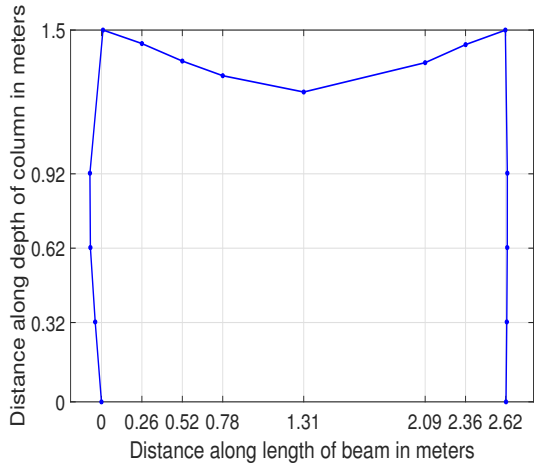


Figure C.1. Normalized mode shape corresponding to the first natural frequency.

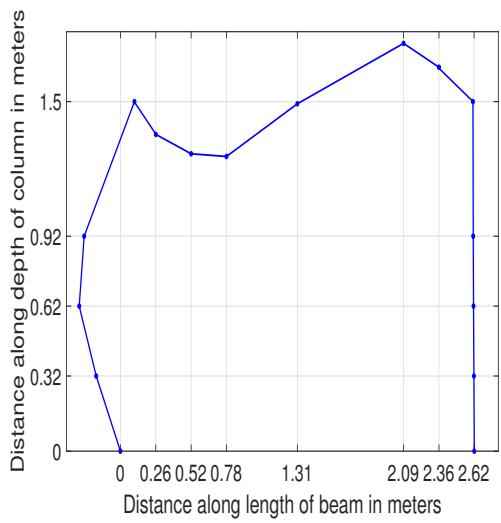


Figure C.2. Normalized mode shape corresponding to the second natural frequency.

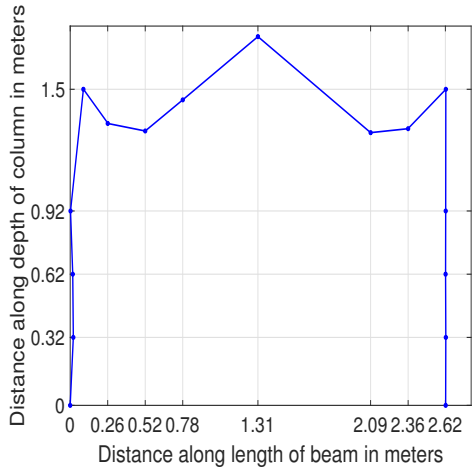


Figure C.3. Normalized mode shape corresponding to the third natural frequency.

D Comparison of virtual damping force for different damage scenarios

The comparison of estimated damping forces corresponding to various damage scenarios both in the time domain and frequency domain is shown in the figures below.

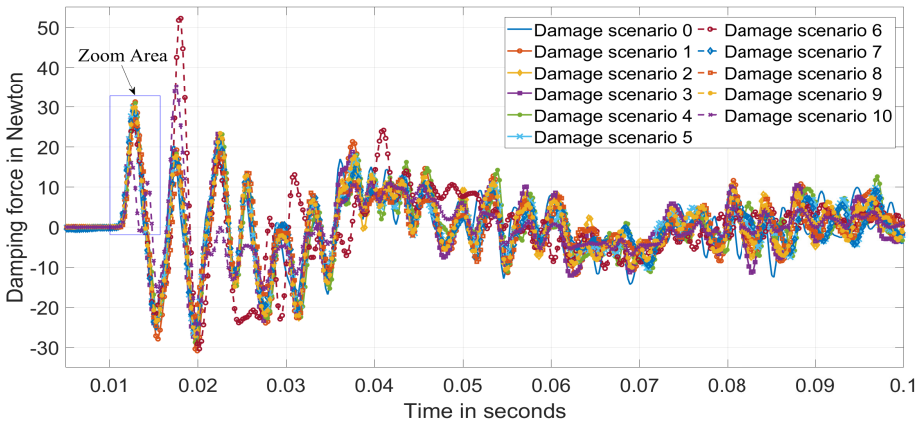


Figure D.1. Estimated damping force for multiple damage scenarios.

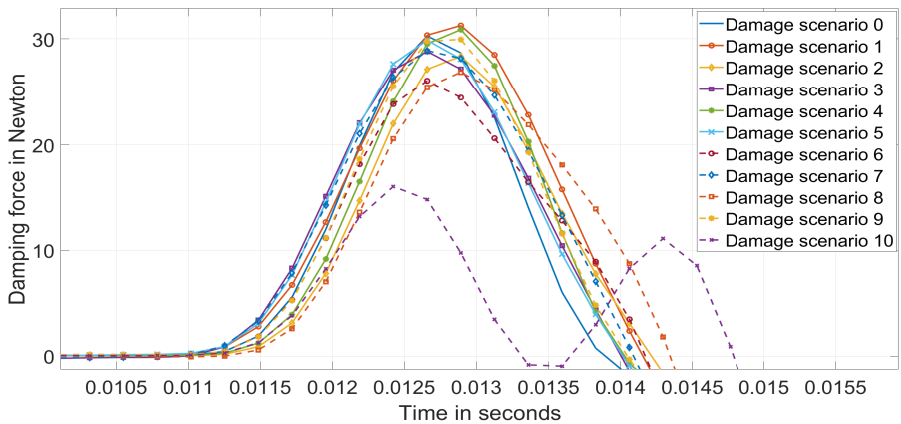


Figure D.2. Estimated damping force for multiple damage scenarios (zoomed).

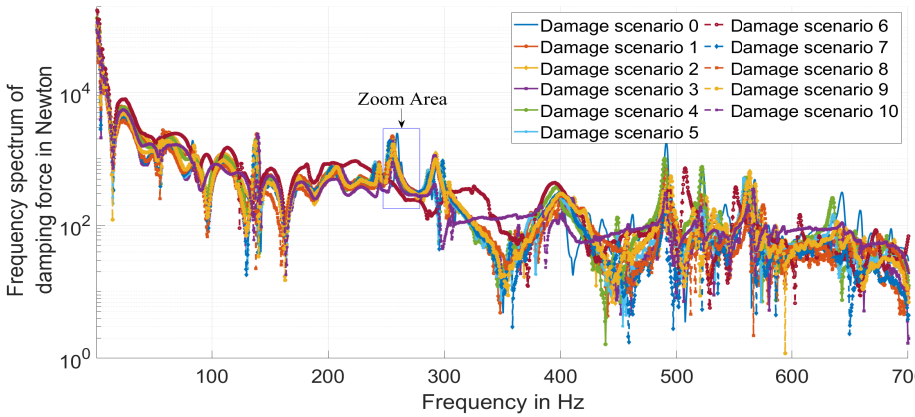


Figure D.3. Frequency spectrum of estimated damping force for multiple damage scenarios.

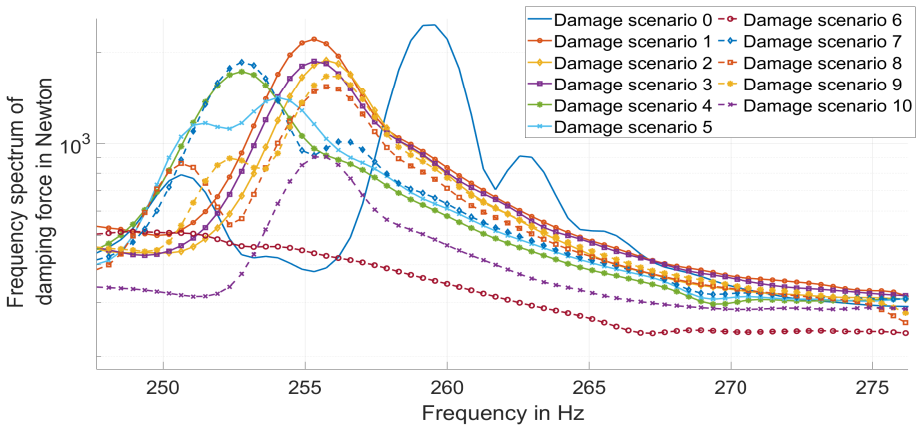


Figure D.4. Frequency spectrum of estimated damping force for multiple damage scenarios (zoomed).

E Comparison of acceleration responses for damage cases 1-10

The estimated acceleration responses for damage scenarios 1 to 10 are plotted against the corresponding measured responses.

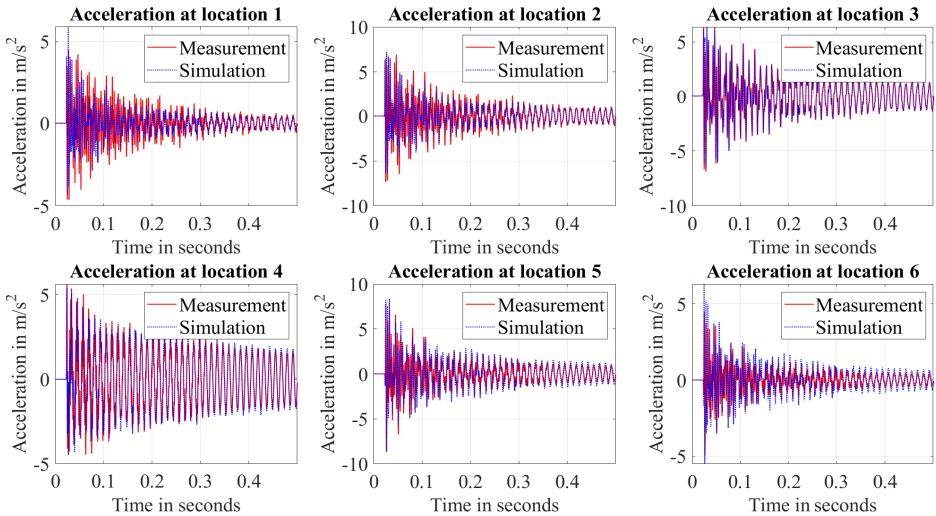


Figure E.1. Comparison of acceleration responses for damage case 1 after damping update.

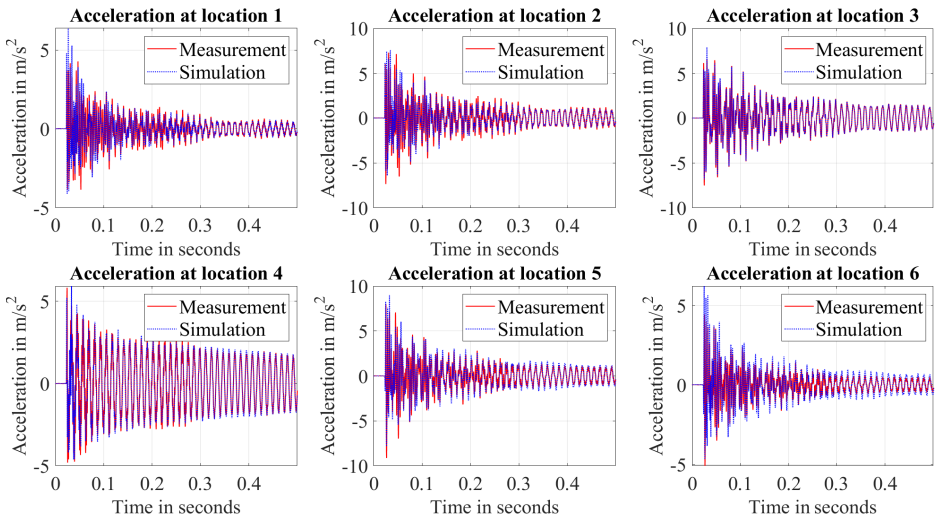


Figure E.2. Comparison of acceleration responses for damage case 2 after damping update.

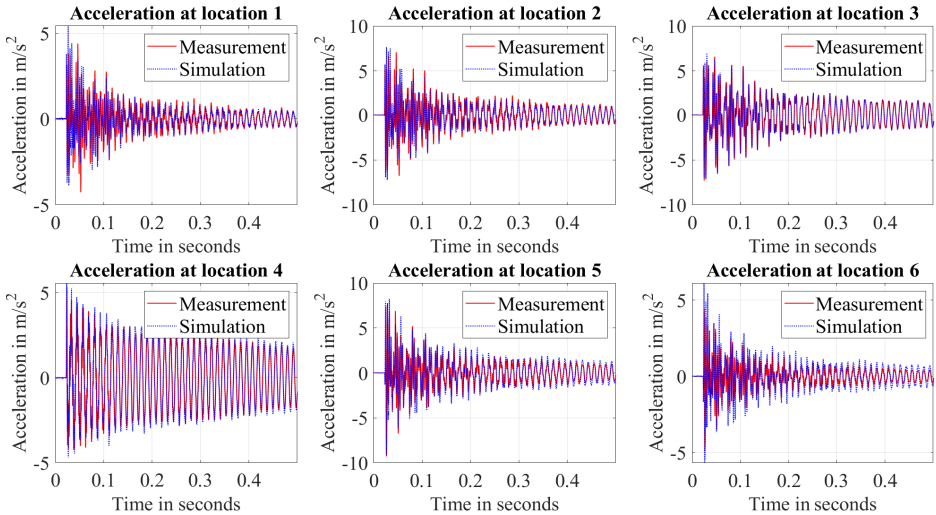


Figure E.3. Comparison of acceleration responses for damage case 3 after damping update.

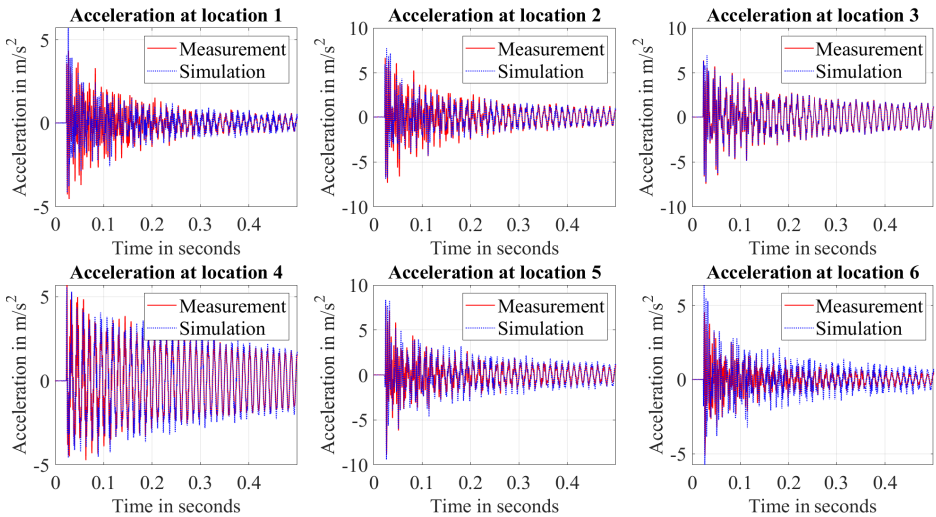


Figure E.4. Comparison of acceleration responses for damage case 4 after damping update.

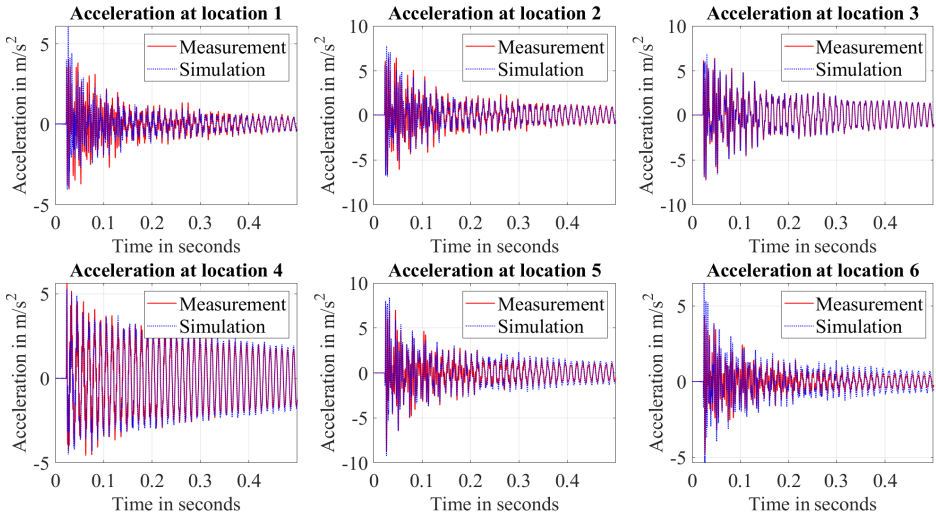


Figure E.5. Comparison of acceleration responses for damage case 5 after damping update.

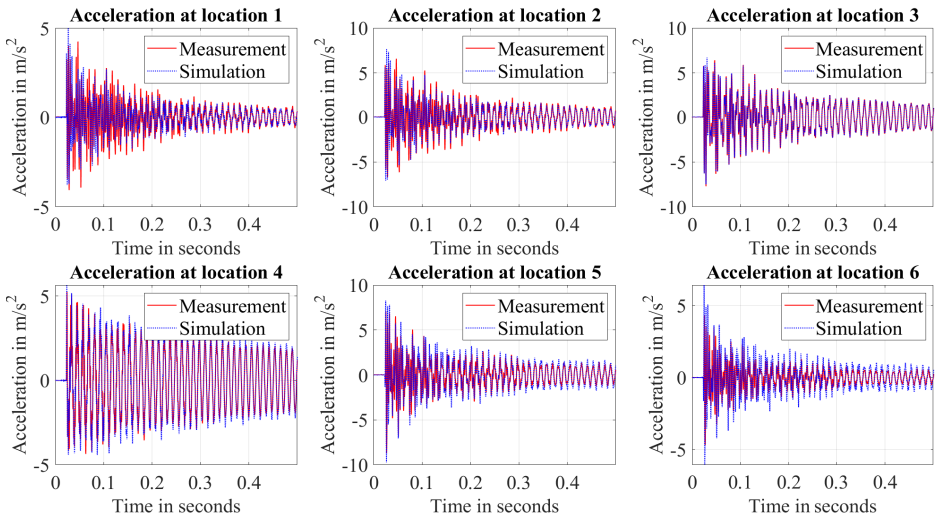


Figure E.6. Comparison of acceleration responses for damage case 6 after damping update.

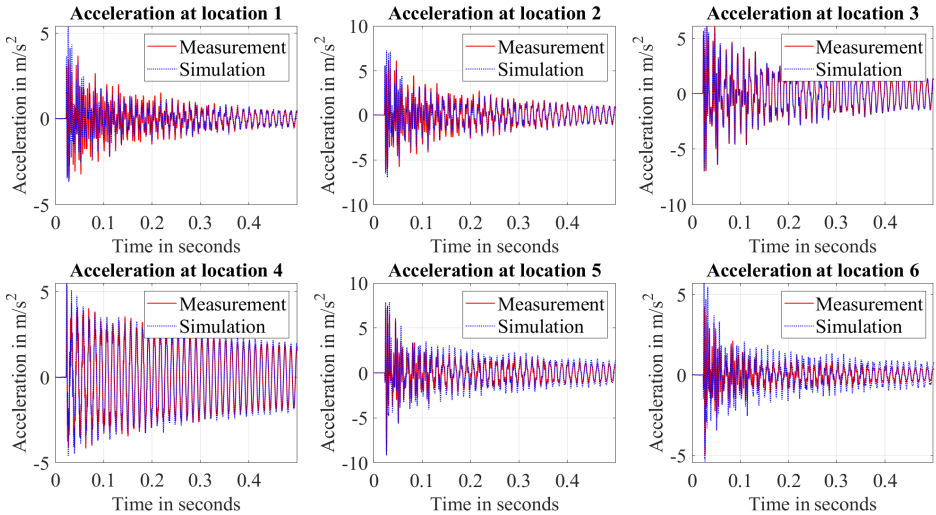


Figure E.7. Comparison of acceleration responses for damage case 7 after damping update.

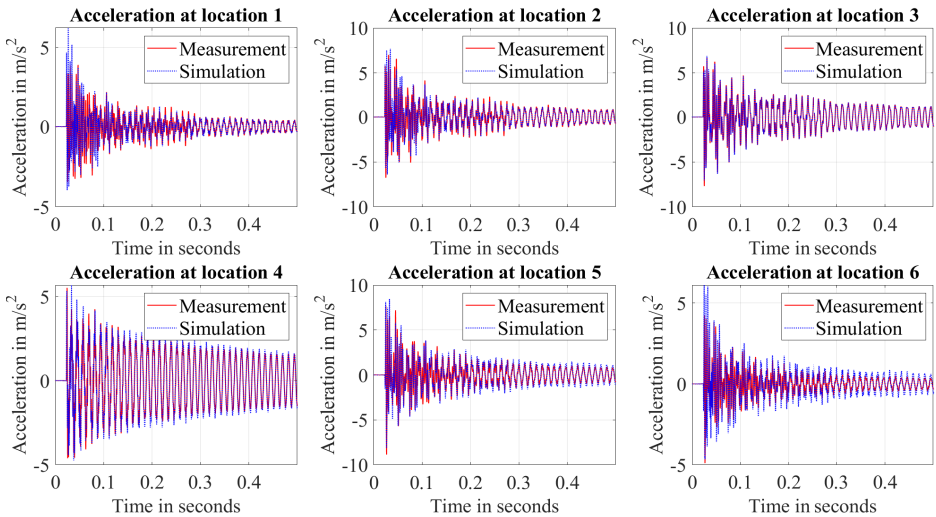


Figure E.8. Comparison of acceleration responses for damage case 8 after damping update.

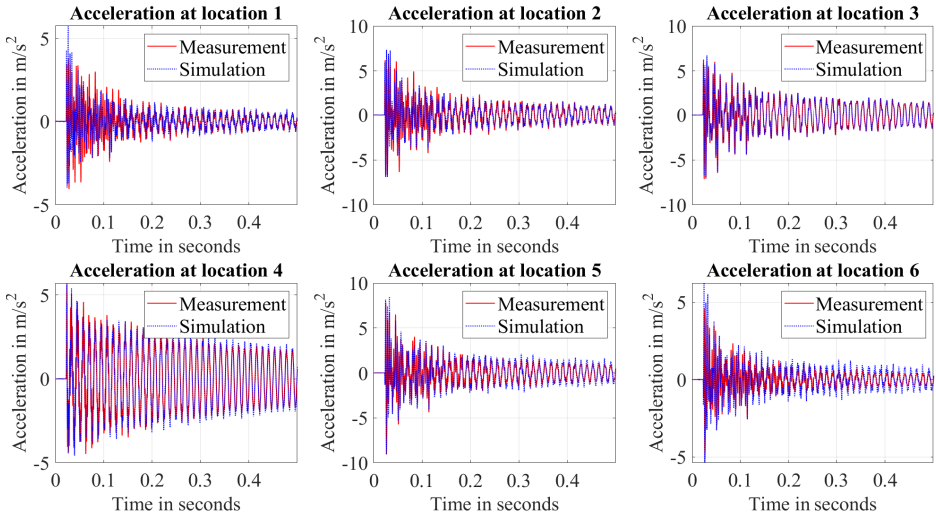


Figure E.9. Comparison of acceleration responses for damage case 9 after damping update.

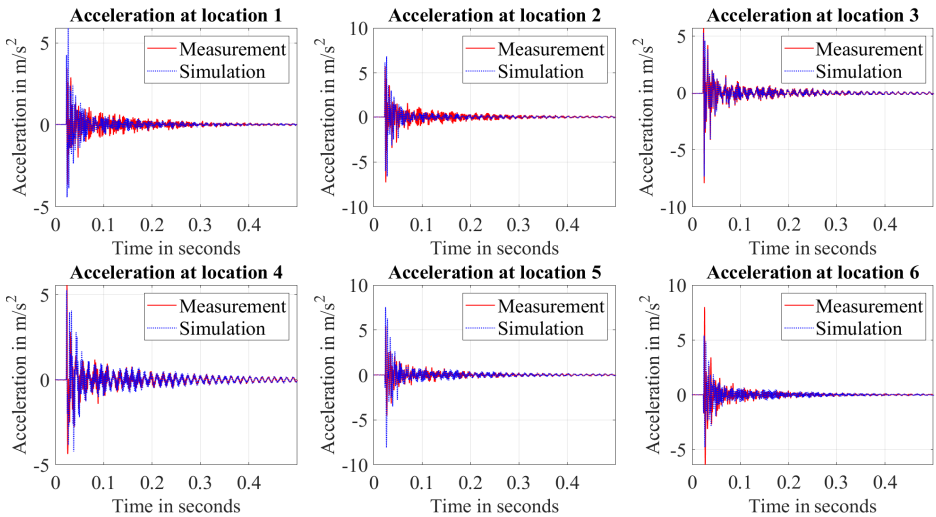


Figure E.10. Comparison of acceleration responses for damage case 10 after damping update.

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